

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: LTZGLUE | Context: Luxembourg Public Administration — Civil Servant NLU Deployment

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Annotation provenance and quality are mixed and weakly tied to civil-administration norms. Sentiment was annotated by two native LTZ speakers with Cohen's Kappa 0.45 — moderate agreement — and disagreements resolved by consensus [Q25, Q26, Q27, Q28]; no documentation of administrative-register expertise. JUDGEWEL NER relies on LLM-as-judge with 'minimal human verification' [Q37]. RTE used ChatGPT-5-mini to verify labels, finding nearly 10% incorrect labels [Q63, Q64], with manual correction afterwards [Q65]. The intent detection translation lacked LTZ register references and could not be systematically verified [Q56]. Authors explicitly caution against 'using benchmark performance as evidence of cultural or demographic coverage' [Q121] and note 'models may reproduce dominant norms while under-representing regional, sociolectal, or multilingual practices' [Q120]. For a deployment where sentiment must capture formal-but-understated frustration in civil correspondence [elicitation Q3, regional context cultural_norms_notes], $\kappa=0.45$ with consumer-facing news comments as source material is insufficient. OC is HIGH priority.

ID	Evidence
Q3	"We evaluate various pre-trained language models for LTZ to present an overview of the current capabilities of these models on the LTZ language." (p.1)
Q25	"In total, we extract 4,583 sentences, which are then annotated by two native speakers of LTZ." (p.3)
Q26	"Annotators are instructed to label each sentence, and to use unsure only when they would otherwise randomly use the other labels." (p.3)
Q27	"We calculated Cohen's Kappa at 0.45." (p.3)
Q28	"For the final set, the annotators agree on a label in cases of label disagreement." (p.3)
Q37	"The resulting sentences are then evaluated using LLMs acting as judges, with minimal human verification to calibrate quality thresholds." (p.4)
Q56	"Due to the lack of LTZ references in this register, it was not possible to systematically verify the translated terminology." (p.5)
Q63	"To achieve (a), we prompted CHATGPT-5-MINI to judge the texts in the improved data and label their quality as either low, medium, or high, keeping only data rated at least medium, removing nearly 25% of the entire dataset (see Appendix 7.5)." (p.5)
Q64	"For (b), we prompted CHATGPT-5-MINI to verify whether the dataset labels remained correct after the first translation and improvement, outputting true or false for each sentence pair (see Appendix 7.6). Nearly 10% of the labels were false." (p.5)
Q65	"We found that the quality improvement step often corrected intentional logical contradictions or factual inaccuracies rather than keeping the original semantics. We therefore adjusted the sentences manually such that they corresponded to the ground truth again, while keeping false positives intact." (p.5)
Q120	"As a result, models may reproduce dominant norms while under-representing regional, sociolectal, or multilingual practices." (p.10)
Q121	"We therefore caution against using benchmark performance as evidence of cultural or demographic coverage." (p.10)

Strengths

- Native LTZ speakers conducted sentiment annotation [Q25] and intent translation [Q52], grounding labels in language-native judgment
- Lothritz et al. NER sub-corpus is fully human-annotated with 'high-precision' result [Q42]
- RTE underwent multi-stage label verification with manual correction [Q63, Q64, Q65]

- Authors transparently disclose annotation limitations [Q113, Q120, Q121]

ID	Evidence
Q25	"In total, we extract 4,583 sentences, which are then annotated by two native speakers of LTZ." (p.3)
Q42	"Annotation was conducted manually, yielding a smaller but high-precision dataset." (p.4)
Q52	"The translations were performed by an LTZ native speaker. In cases of uncertainty, additional native LTZ speakers were consulted." (p.4)
Q63	"To achieve (a), we prompted CHATGPT-5-MINI to judge the texts in the improved data and label their quality as either low, medium, or high, keeping only data rated at least medium, removing nearly 25% of the entire dataset (see Appendix 7.5)." (p.5)
Q64	"For (b), we prompted CHATGPT-5-MINI to verify whether the dataset labels remained correct after the first translation and improvement, outputting true or false for each sentence pair (see Appendix 7.6). Nearly 10% of the labels were false." (p.5)
Q65	"We found that the quality improvement step often corrected intentional logical contradictions or factual inaccuracies rather than keeping the original semantics. We therefore adjusted the sentences manually such that they corresponded to the ground truth again, while keeping false positives intact." (p.5)
Q113	"Although we draw on established GLUE-style tasks, some annotation decisions and class distributions are necessarily influenced by resource constraints. Certain tasks exhibit label imbalance or rely on automatic preprocessing, which may introduce biases that we cannot fully quantify." (p.9)
Q120	"As a result, models may reproduce dominant norms while under-representing regional, sociolectal, or multilingual practices." (p.10)
Q121	"We therefore caution against using benchmark performance as evidence of cultural or demographic coverage." (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: INSUFFICIENT DOCUMENTATION on whether labels reflect civil-administration stakeholder perspectives. Sentiment annotators are described only as native LTZ speakers [Q25]; no administrative or civil-service background is documented.

OC-2: Likely disagreement risk is high for sentiment/urgency: regional culture features formal, understated expression of frustration [regional context cultural_norms_notes]; news-comment-trained annotators with $\kappa=0.45$ [Q27] are unlikely to encode that register reliably.

OC-3: Documentation is partial — annotator counts and native-speaker status reported [Q25, Q52], but no demographic, professional, or domain-expertise breakdown provided. No Datasheet/Data Statement is referenced.

OC-4: Re-annotation by civil-servant or administration-experienced annotators is recommended; the elicitation already provisions a real-time civil-servant feedback loop [elicitation Q3] but pre-deployment re-annotation would be additionally valuable.

OC-5: Sentiment disagreements were resolved by consensus between two annotators [Q28], which can erase minority perspectives; this is a risk for a population with diverse linguistic backgrounds (frontaliers, non-native LTZ writers).

OC-6: Convergent validity is at risk because labels may not correlate with civil-servant judgments of administrative correspondence; external validity is at risk because the source distribution (RTL comments, dictionary, voice-assistant) differs from citizen-administration correspondence [Q112, Q119].

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Q3	"We evaluate various pre-trained language models for LTZ to present an overview of the current capabilities of these models on the LTZ language." (p.1)
Q25	"In total, we extract 4,583 sentences, which are then annotated by two native speakers of LTZ." (p.3)
Q27	"We calculated Cohen's Kappa at 0.45." (p.3)
Q28	"For the final set, the annotators agree on a label in cases of label disagreement." (p.3)
Q52	"The translations were performed by an LTZ native speaker. In cases of uncertainty, additional native LTZ speakers were consulted." (p.4)
Q112	"Coverage across domains, registers, and demographic varieties may also be limited. LTZ displays substantial orthographic and sociolinguistic variation, yet most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts." (p.9)
Q119	"LTZ is a small language community, and linguistic data often originate from a limited set of public domains." (p.10)
Q120	"As a result, models may reproduce dominant norms while under-representing regional, sociolectal, or multilingual practices." (p.10)
Q121	"We therefore caution against using benchmark performance as evidence of cultural or demographic coverage." (p.10)

Information Gaps

- Annotator demographics, professional background, civil-administration experience
- Whether sentiment label boundaries were validated against civil-correspondence-specific examples
- Granular agreement statistics by class (e.g., neutral vs. negative) to assess understated-frustration handling

Requires Expert Verification

- Civil-servant review of a held-out sample of sentiment labels for civil-correspondence relevance
- Re-annotation of a small evaluation set by administrative-domain annotators to estimate convergent validity

Remediation

Gap 1: Sentiment annotators lack documented administrative-register expertise; $\kappa=0.45$ is moderate; LLM-assisted labeling introduces uncertainty in RTE and JUDGEWEL

Recommendation: Re-annotate a calibration sample of sentiment and any new routing categories with civil servants or administration-experienced annotators; report per-class agreement and disagreement patterns; use the deployment's human-in-the-loop feedback to drive ongoing label refinement.